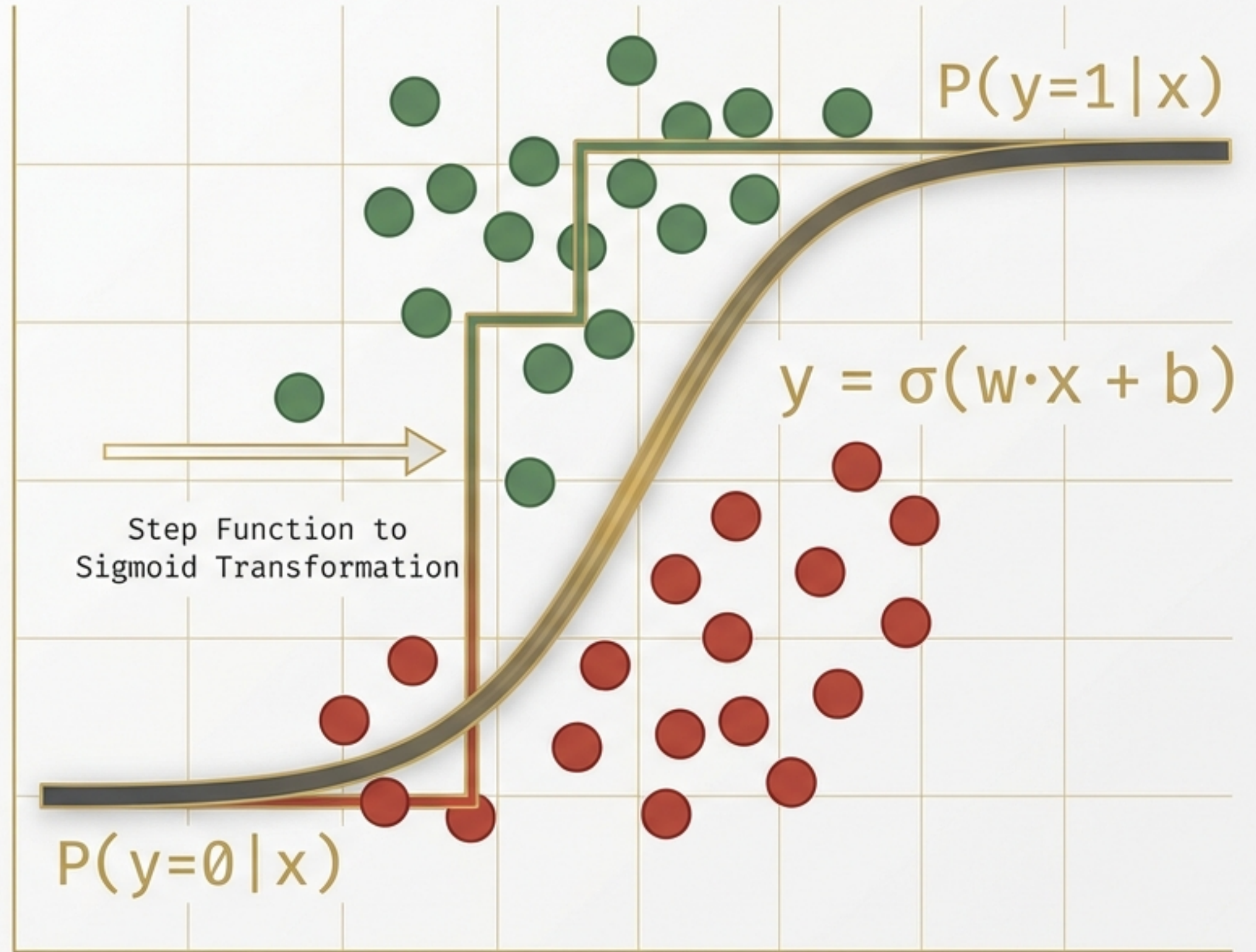


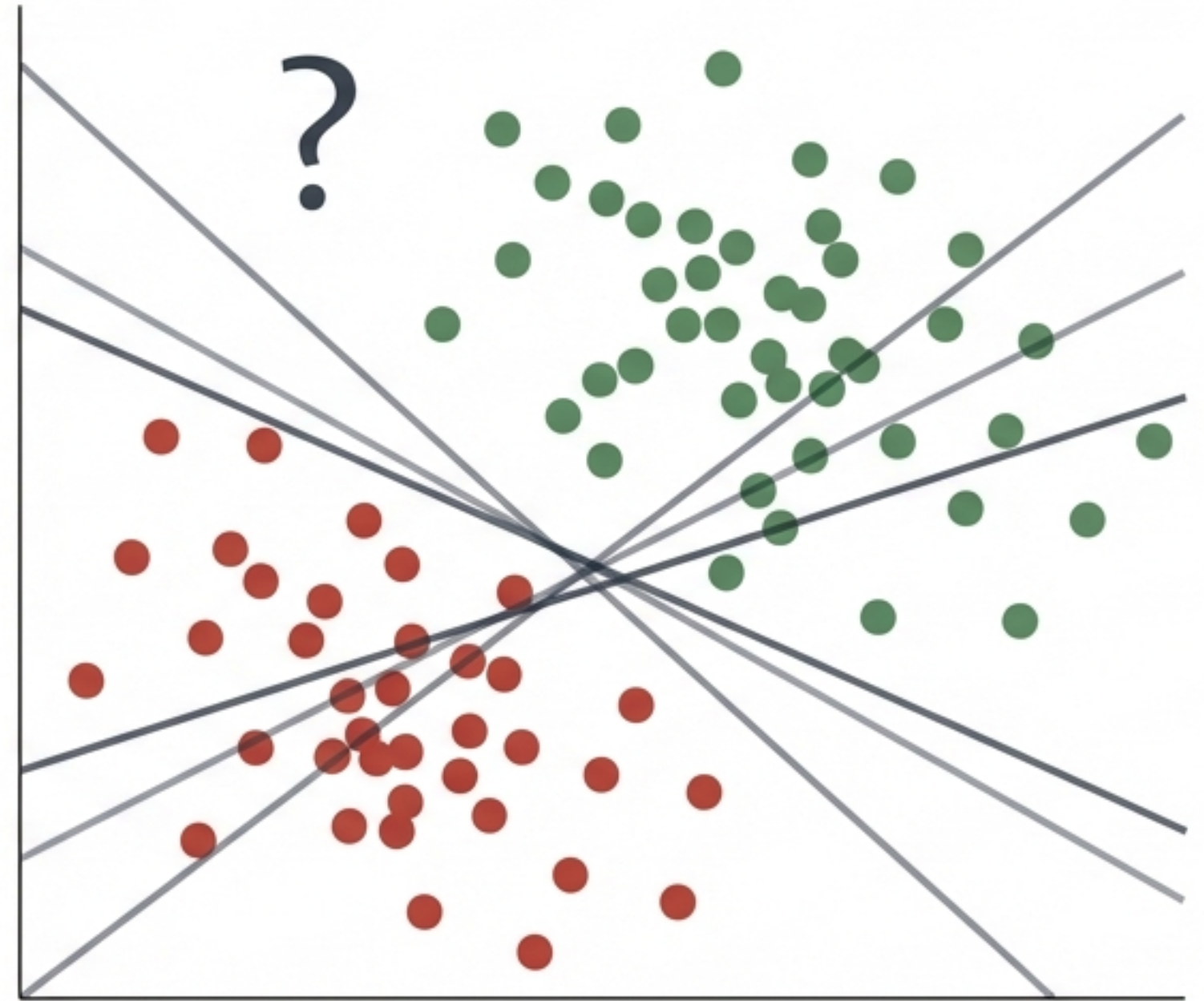
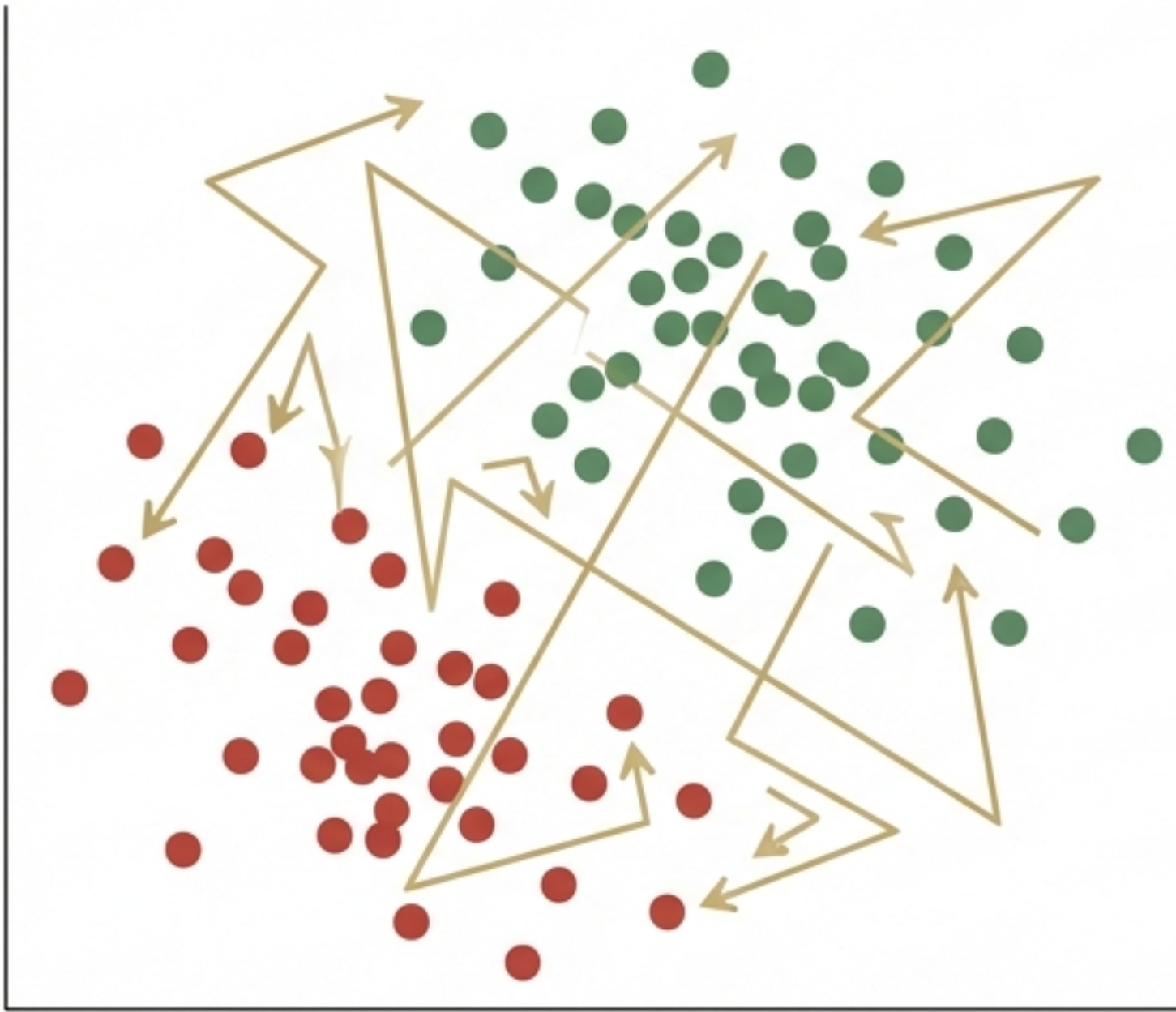
The Road to Binary Cross Entropy

- Deriving the core loss function of Logistic Regression from first principles.
- Moving from visual intuition to mathematical certainty.



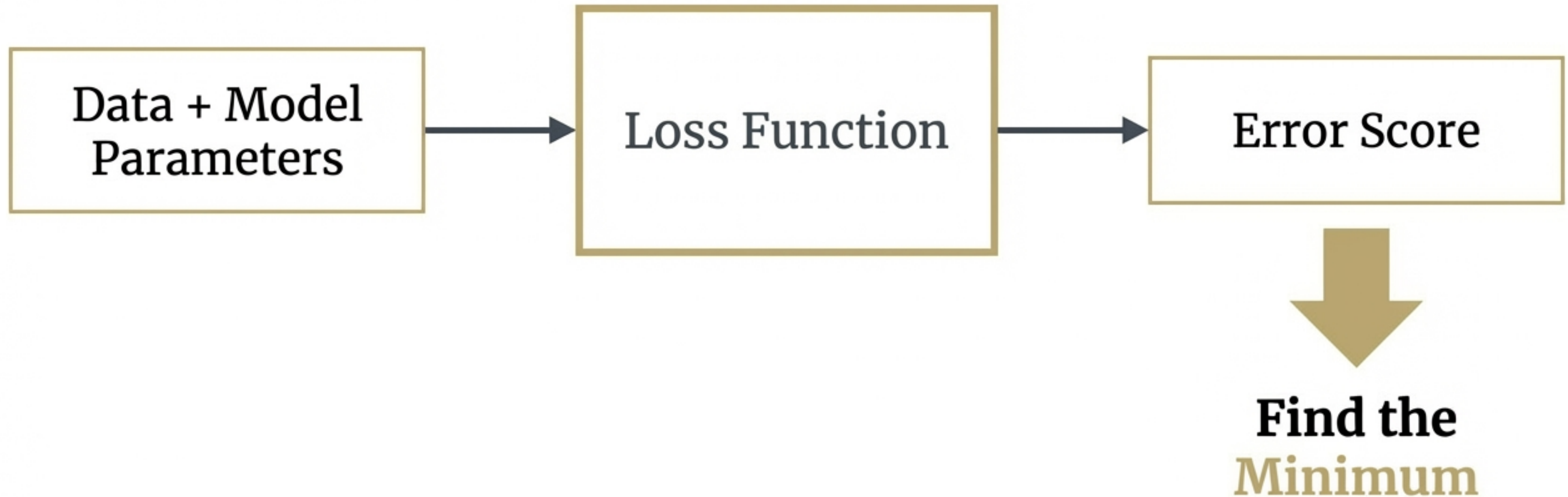
The limitation of random adjustments

Adjusting a model solely based on misclassified points is an endless guessing game. Without a universal metric, we cannot **guarantee convergence** on the **absolute best line**. We need a mathematically rigorous way to score our model.



Finding the mathematical baseline

- True machine learning relies on defining a Loss Function (or Error Function).
- This function continuously quantifies exactly how much our model is failing.
- Once we define the error, our sole objective is to minimise it.



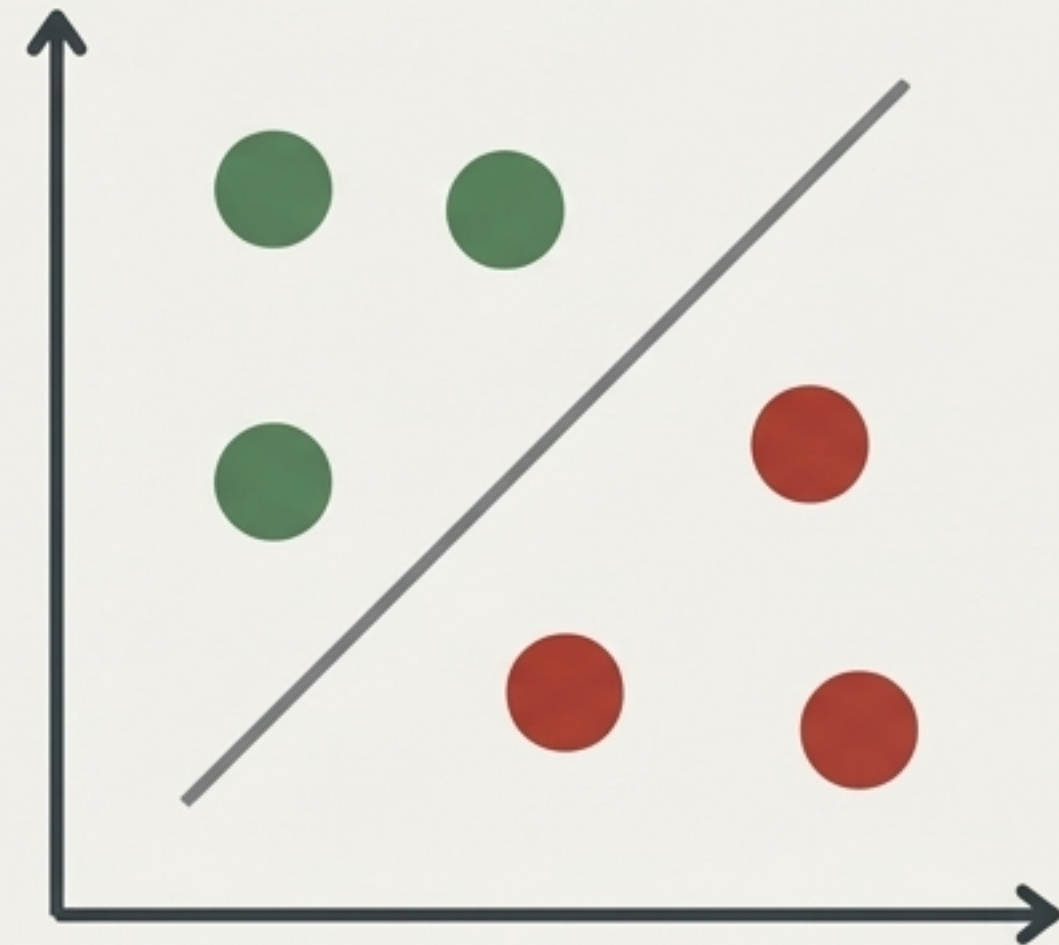
The visual intuition test

- Human intuition instantly identifies Model 2 as the superior classifier.
- The challenge: How do we teach a machine to reach this exact same conclusion?

Model 1

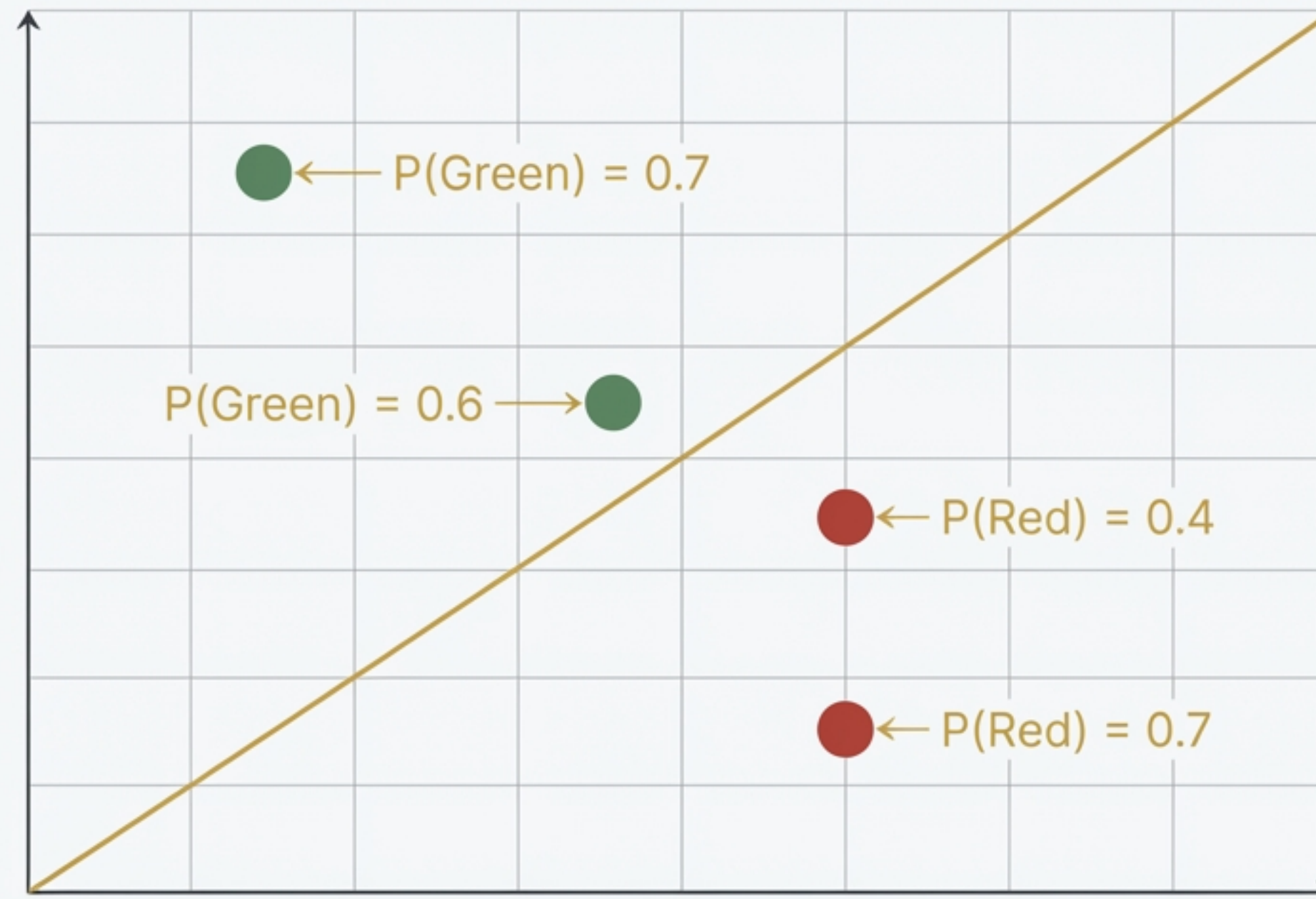


Model 2



Translating visual boundaries into probabilities

- We use the sigmoid function to calculate $\hat{y}$ — the predicted probability for every point.
- The closer a point is to the correct side of the boundary, the higher its probability score.



Scoring models using Maximum Likelihood

- By multiplying the predicted probabilities of the correct classes together, we get a total score.
- Model 2 yields a higher product. The math finally aligns with our visual intuition.

Model 1 Math:

$$0.7 \times 0.4 \times 0.4 \times 0.8 = 0.0896$$

Model 2 Math:

$$0.7 \times 0.6 \times 0.4 \times 0.7 = 0.1176$$



The fatal flaw of multiplying probabilities

- Real-world datasets contain tens of thousands of rows.
- Multiplying thousands of fractions inevitably results in an infinitesimally small number.
- Computers cannot process this without rounding down to zero (Underflow).

$$0.7 \times 0.6 \times 0.2 \times \dots \times (10,000 \text{ times}) = 0.00000000000000000000\dots$$

The Logarithmic workaround

- Logarithms allow us to transform complex products into manageable summations.
- As new data points arrive, we simply calculate the log of their prediction and add it to the running total.

$$\log(A \times B) = \log(A) + \log(B)$$

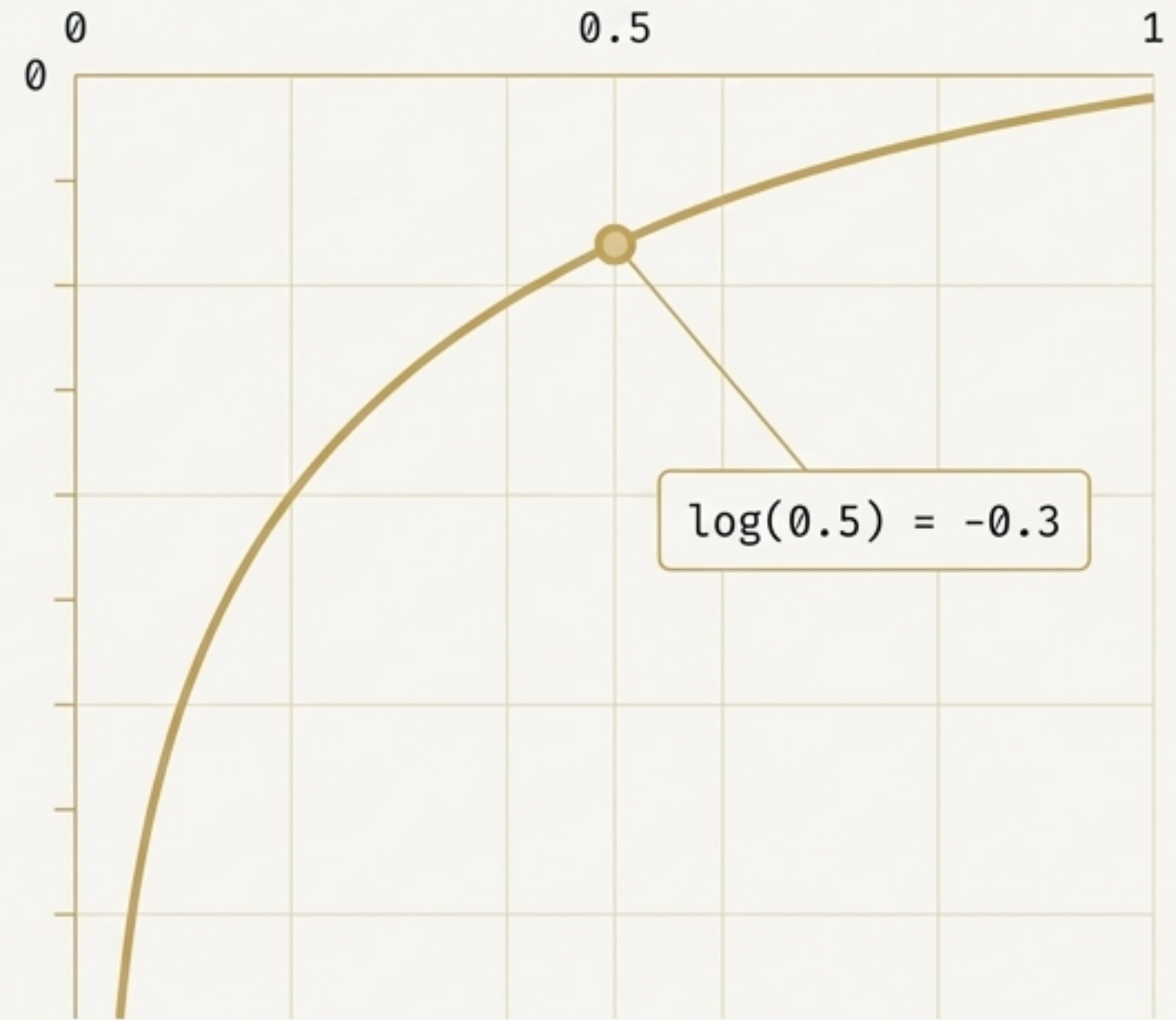


$$\log(0.7) + \log(0.6) + \log(0.4) + \log(0.7)$$

The negative number problem

- Because probability always sits between 0 and 1, taking the log always returns a negative value.
- Adding up thousands of negative numbers creates an unreadable, deeply negative error score.

$$-0.15 + (-0.22) + (-0.40) + (-0.15) = -0.92$$



The Cross-Entropy paradigm shift

- By simply multiplying the entire summation by -1, we convert negative scores into positive ones.
- This forces a shift in our objective: we move from trying to maximise likelihood to minimising Cross-Entropy.

$$- \left[\log(p_1) + \log(p_2) + \dots \right]$$

Maximum Likelihood



Goal: Maximise the product of probabilities.

Cross-Entropy



Goal: Minimise the sum of negative logs.

The challenge of drafting a universal formula

- We cannot use a simplistic formula because Green points and Red points require calculating opposite probabilities.
- We need a single, elegant mathematical expression that handles both cases simultaneously without IF/ELSE statements.

If point is Green ($y=1$):
Use $-\log(\hat{y})$

If point is Red ($y=0$):
Use $-\log(1-\hat{y})$

The Mathematical Switch

- This structure acts as an automatic, built-in switch.
- Depending on the true label (y), it perfectly preserves the relevant calculation while multiplying the irrelevant part by zero.

$$- [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

Proving the switch in action

- The formula elegantly collapses to fit the exact probability requirement of the data point.

When $y = 1$

$$- [1 \cdot \log(\hat{y}) + \cancel{(1-1) \cdot \log(1-\hat{y})}] \longrightarrow -\log(\hat{y})$$

When $y = 0$

$$- [\cancel{0 \cdot \log(\hat{y})} + (1-0) \cdot \log(1-\hat{y})] \longrightarrow -\log(1-\hat{y})$$

Binary Cross Entropy (Log Loss)

- We have successfully transformed visual intuition into the fundamental loss function of Logistic Regression.
- Every variable solves a specific scaling, calculation, or logic problem.

The diagram shows the formula for Binary Cross Entropy (Log Loss) with three annotations:

- An arrow points from the text "Average over n points" to the $\frac{1}{n}$ term in the formula.
- An arrow points from the text "Negative inversion" to the minus sign in the formula.
- An arrow points from the text "The mathematical switch" to the $(1 - y_i)$ term in the formula.

$$L = -\frac{1}{n} \sum [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

The next step: Finding the minimum

- Unlike Linear Regression, Binary Cross Entropy has no closed-form solution. We cannot solve it in a single calculation.
- To find the precise coefficients that minimise this error, we must apply Gradient Descent.

